# Diagnostic performance of cone-beam computed tomography for scaphoid fractures—A systematic review and diagnostic meta-analysis


## Authors

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## Supplementary Online Content


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**Table 1. Search strategy**

| Recent queries in PubMed on March 25, 2020 | | |
|---|---|---|
| **Search** | **Query** | **Items found** |
| 1 | Search cone beam | 15492 |
| 2 | Search cone-beam | 14773 |
| 3 | Search scaphoid | 5028 |
| 4 | Search "scaphoid bone"[MeSH Terms] | 2185 |
| 5 | Search wrist | 43688 |
| 6 | Search "wrist joint"[MeSH Terms] | 10087 |
| 7 | Search "wrist"[MeSH Terms] | 8870 |
| 8 | Search radiocarpal | 1455 |
| 9 | Search fracture* | 301881 |
| 10 | Search fracture | 279410 |
| 11 | Search "fractures, bone"[MeSH Terms] | 181784 |
| 12 | Search CBCT | 8550 |
| 13 | Search computed tomography | 578061 |
| 14 | Search "tomography, x-ray computed"[MeSH Terms] | 424374 |
| 15 | Search ((cone beam) OR cone-beam) OR CBCT | 16002 |
| 16 | Search (computed tomography) OR "tomography, x-ray computed"[MeSH Terms] | 578061 |
| 17 | Search (((computed tomography) OR "tomography, x-ray computed"[MeSH Terms])) AND (((cone beam) OR cone-beam) OR CBCT) | 14059 |
| 18 | Search (((((scaphoid) OR "scaphoid bone"[MeSH Terms]) OR wrist) OR "wrist joint"[MeSH Terms]) OR "wrist"[MeSH Terms]) OR radiocarpal | 45958 |
| 19 | Search ((fracture*) OR fracture) OR "fractures, bone"[MeSH Terms] | 304052 |
| 20 | Search (((((((scaphoid) OR "scaphoid bone"[MeSH Terms]) OR wrist) OR "wrist joint"[MeSH Terms]) OR "wrist"[MeSH Terms]) OR radiocarpal)) AND (((fracture*) OR fracture) OR "fractures, bone"[MeSH Terms]) | 10553 |
| 21 | Search (((((((((scaphoid) OR "scaphoid bone"[MeSH Terms]) OR wrist) OR "wrist joint"[MeSH Terms]) OR "wrist"[MeSH Terms]) OR radiocarpal)) AND (((fracture*) OR fracture) OR "fractures, bone"[MeSH Terms]))) AND ((((computed | 17 |

| | | |
|---|---|---|
| | tomography) OR "tomography, x-ray computed"[MeSH Terms])) AND (((cone beam) OR cone-beam) OR CBCT)) | |
| **Recent queries in Web of Science on March 25, 2020** | | |
| **Search** | **Query** | **Items found** |
| 1 | TOPIC: (cone beam) | 16,517 |
| 2 | TOPIC: (cone-beam) | 13,476 |
| 3 | TOPIC: (CBCT) | 7,871 |
| 4 | TOPIC: (computed tomography) | 261,607 |
| 5 | TOPIC: (wrist) | 31,555 |
| 6 | TOPIC: (scaphoid) | 2,941 |
| 7 | TOPIC: (radiocarpal) | 949 |
| 8 | TOPIC: (fracture*) | 405,128 |
| 9 | #3 OR #2 OR #1 | 18,122 |
| 10 | #9 AND #4 | 9,986 |
| 11 | #7 OR #6 OR #5 | 33,207 |
| 12 | #11 AND #8 | 6,050 |
| 13 | #12 AND #10 | 19 |
| **Recent queries in Scopus on March 25, 2020** | | |
| **Search** | **Query** | **Items found** |
| 1 | TITLE-ABS-KEY (cone AND beam) | 27344 |
| 2 | TITLE-ABS-KEY (cone-beam) | 22163 |
| 3 | TITLE-ABS-KEY (cbct) | 11381 |
| 4 | TITLE-ABS-KEY (computed AND tomography) | 642319 |
| 5 | (TITLE-ABS-KEY (cone AND beam) OR TITLE-ABS-KEY (cone-beam) OR TITLE-ABS-KEY (cbct)) | 28439 |
| 6 | TITLE-ABS-KEY (computed AND tomography) AND (TITLE-ABS-KEY (cone AND beam) OR TITLE-ABS-KEY (cone-beam) OR TITLE-ABS-KEY (cbct)) | 18983 |
| 7 | TITLE-ABS-KEY (scaphoid) | 6707 |
| 8 | TITLE-ABS-KEY (wrist) | 65263 |
| 9 | TITLE-ABS-KEY (radiocarpal) | 1633 |
| 10 | TITLE-ABS-KEY (fracture*) | 789632 |
| 11 | TITLE-ABS-KEY (scaphoid) OR TITLE-ABS-KEY (wrist) OR TITLE-ABS-KEY (radiocarpal) | 68158 |

| | | |
|---|---|---|
| 12 | (TITLE-ABS-KEY (scaphoid) OR TITLE-ABS-KEY (wrist) OR TITLE-ABS-KEY (radiocarpal)) AND TITLE-ABS-KEY (fracture*) | 14375 |
| 13 | ( ( TITLE-ABS-KEY ( scaphoid ) OR TITLE-ABS-KEY (wrist) OR TITLE-ABS-KEY (radiocarpal )) AND TITLE-ABS-KEY( fracture*)) AND ( TITLE-ABS-KEY (computed AND tomography) AND (TITLE-ABS-KEY ( cone AND beam) OR TITLE-ABS-KEY (cone-beam) OR TITLE-ABS-KEY (cbct))) | 23 |
| **Recent queries in Embase on March 25, 2020** | | |
| **Search** | **Query** | **Items found** |
| 1 | cone AND beam | 21654 |
| 2 | 'cone beam' | 20636 |
| 3 | cbct | 12570 |
| 4 | computed AND tomography | 504777 |
| 5 | #1 OR #2 OR #3 | 23924 |
| 6 | #4 AND #5 | 20006 |
| 7 | scaphoid | 6463 |
| 8 | wrist | 62364 |
| 9 | radiocarpal | 1922 |
| 10 | fracture* | 404642 |
| 11 | #7 OR #8 OR #9 | 65219 |
| 12 | #10 AND #11 | 14107 |
| 13 | #6 AND #12 | 24 |
| **Recent queries in Cochrane Library on March 25, 2020** | | |
| **Search** | **Query** | **Items found** |
| 1 | cone beam | 873 |
| 2 | cone-beam | 853 |
| 3 | cbct | 656 |
| 4 | computed tomography | 15255 |
| 5 | #1 or #2 | 873 |
| 6 | #5 and #4 | 737 |
| 7 | #6 or #3 | 1031 |
| 8 | scaphoid | 197 |
| 9 | wrist | 5460 |

| | | |
|---|---|---|
| 10 | radiocarpal | 49 |
| 11 | fracture* | 22012 |
| 12 | MeSH descriptor: [Fractures, Bone] explode all trees | 5777 |
| 13 | #8 or #9 or # 10 | 936641 |
| 14 | #11 or #12 | 22036 |
| 15 | #13 and #14 | 14499 |
| 16 | #7 and #15 | 44 |

**Table 2. Summary performance estimates of diagnostic parameters**

| Parameter | Point estimate | 95% CI |
|---|---|---|
| Sensitivity | 0.88 | 0.74-0.95 |
| Specificity | 0.99 | 0.93-1.00 |
| PLR | 119.0 | 11.7-1210.2 |
| NLR | 0.12 | 0.06-0.27 |
| DOR | 957 | 110-8346 |

Footnote: CI, confidence interval; DOR, diagnostic odds ratio; NLR, negative likelihood ratio; PLR, positive likelihood ratio

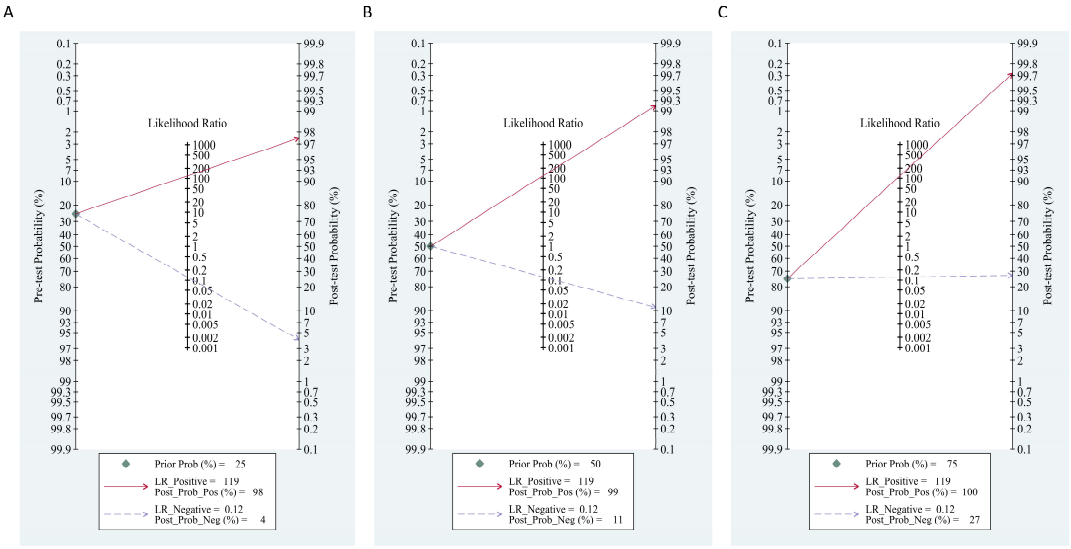


**Figure 1. Analysis of Fagan plot to evaluate the clinical utility of cone-beam computed tomography for detecting scaphoid fracture.** (A) Pretest probability of 25% for scaphoid fractures. (B) Pretest probability of 50% for scaphoid fractures. (C) Pretest probability of 75% for scaphoid fractures. Fagan plot consists of a vertical axis on the left showing pretest probability, an axis in the middle representing the likelihood ratio, and a vertical axis on the right showing posttest probability. NLR, negative likelihood ratio; PLR, positive likelihood ratio.